

Package #171

Embed a shape-memory alloy lattice within the wall of each catheter lumen that self-expands to a preset diameter after insertion, creating an internal scaffold that resists lumen narrowing while keeping the catheter's external profile unchanged.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'lumen occlusion' concerning multi-lumen drainage catheter. User-submitted engineering problem (session input): "Why do multi-lumen drainage catheters lose flow, and would a narrower lumen resist occlusion? (baseline comparison)". [fda_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2023-2026). Illustrative records: 'Is Fibrin a Suitable Matrix for Small-Diameter Bioartificial Vascular Grafts? An In Vitro Short-Term Hemocompatibility Study.'; 'Urinary catheters: state of the art and future perspectives - a narrative review.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Self-expanding metal lattice (as used in aerospace deployable structures) exerts a constant outward radial force that preserves conduit diameter under external pressure or internal blockage.

Contents:

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Status: ENGINEERING_DEFINITION. Not physically validated.